CHAPTER 5 : Hydrocarbons

PSPM 2011 / 2012

Q. 1 (a) Based on the reaction scheme below: 8 m

$$CH_3CH_2CH_2CH_3 \xrightarrow{\textbf{\textit{A}}} \textbf{\textit{B}} \text{ (major product)}$$

$$\textbf{\textit{B}} \xrightarrow{\textbf{\textit{C}}} \underset{\displaystyle CH_3CHCH_2CH_3}{\overset{\displaystyle OH}{|}}$$

(i) Give reagents ***A*** and ***C***, and the structure of ***B***.

(ii) Show initiation, propagation and termination steps in the mechanism for the formation of ***B***.

Answer (a)

(i) A = Cl_2, uv or Br_2, uv

B = $CH_3CH(Cl)CH_2CH_3$ or $CH_3CH(Br)CH_2CH_3$

C = aqueous NaOH or KOH

(ii) <u>Initiation step</u>

X_2 = Cl_2 or Br_2

$$X{-}X \xrightarrow{uv} 2X\bullet$$

<u>Propagation Step</u>

$$CH_3CH_2\underset{\displaystyle CH_3}{\underset{|}{HC}}{-}H + X\bullet \longrightarrow CH_3CH_2\dot{C}HCH_3 + HX$$

$$CH_3CH_2\dot{C}HCH_3 + X{-}X \longrightarrow CH_3CH_2\overset{\displaystyle X}{\overset{|}{C}}HCH_3 + X\bullet$$

<u>Termination Step</u>

$$\bullet X + X\bullet \longrightarrow X_2$$

$$CH_3CH_2\dot{C}HCH_3 + X\bullet \longrightarrow CH_3CH_2\overset{\displaystyle X}{\overset{|}{C}}HCH_3$$

Q. 1 (b) Acid-catalysed dehydration of 1-pentanol gives a mixture of alkenes. 7 m

(i) Draw the structures of **all** the products formed, and label the major product.

(ii) State the rule used to determine the major product.

(iii) Write the chemical equations when the minor product reacts with chlorine gas; acidified water; and bromine water.

Answer (b)

(i) $CH_3CH_2CH_2CH{=}CH_2$ + $CH_3CH_2CH{=}CHCH_3$
(major)

(ii) Saytzeff Rule

(iii) $CH_3CH_2CH_2CH{=}CH_2 + Cl_2 \longrightarrow CH_3CH_2CH_2CH(Cl)CH_2(Cl)$

$CH_3CH_2CH_2CH{=}CH_2 + H_3O^+$ or $H_2O/H^+ \longrightarrow CH_3CH_2CH_2CH(OH)CH_3$

$CH_3CH_2CH_2CH{=}CH_2 + Br_2, H_2O \longrightarrow CH_3CH_2CH_2CH(OH)CH_2(Br)$
or $Br_2(aq)$

Q. 2 2-methylpentane reacts with Br_2 in the presence of light to yield ***L***, a monobromination product. Dehydrohalogenation of compound ***L*** forms ***M*** and ***N*** as the respective major and minor products. Oxidative cleavage on ***M*** gives ketone ***P*** and carboxylic acid ***Q***; whereas similar reaction on ***N*** forms ketone ***R*** and CO_2. Treatment of ***M*** with acidified water yields ***S***. 20 m

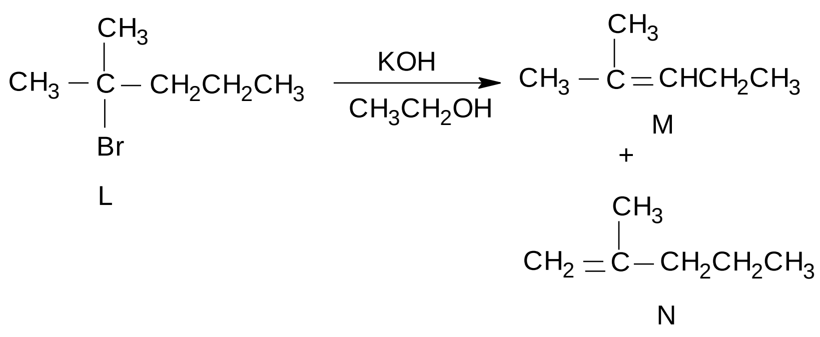
Write the chemical equations for complete combustion of 2-methylpentane; dehydrohalogenation of ***L***; and oxidative cleavage of ***M***.

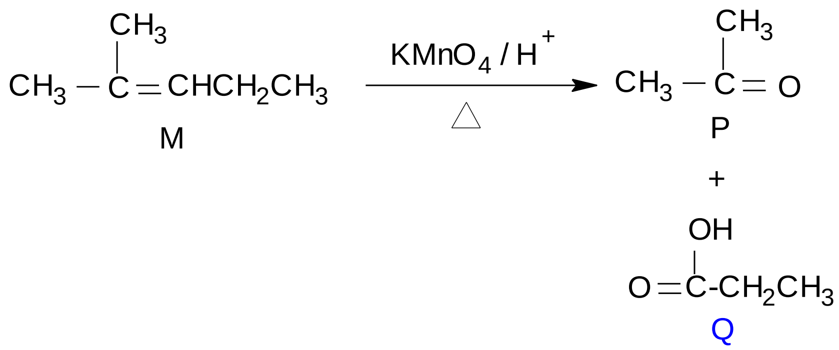
Suggest the structures of ***L***, ***M***, ***N***, ***P***, ***Q***, ***R*** and ***S***. Write the mechanism for the formation of ***S***.

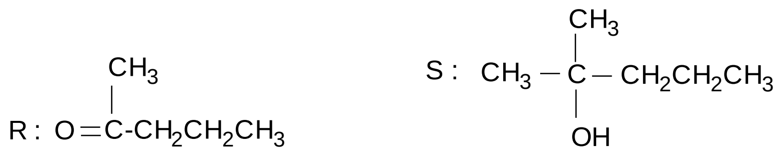
Show how ***Q*** can be synthesized from ethane.

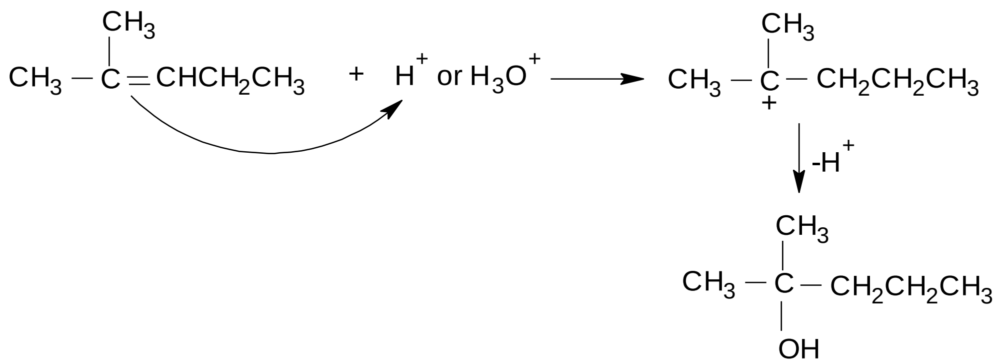
Answer

$C_6\,H_4 + 19/2\ O_2 \rightarrow 6\ CO_2 + 7\ H_2O$









$$CH_3-CH_3 \xrightarrow[uv]{X_2} CH_3-CH_2X \xrightarrow[KCN]{NaCN\ or} CH_3-CH_2CN \xrightarrow{H_3O^+} \underset{Q}{CH_3-CH_2COOH}$$

OR

$$CH_3-CH_3 \xrightarrow[uv]{X_2} CH_3-CH_2X \xrightarrow[Ether]{Mg} CH_3-CH_2MgX \xrightarrow[ii)\ H_3O^+]{i)\ CO_2} \underset{Q}{CH_3-CH_2COOH}$$

PSPM 2012 / 2013

Q. 3 (a) Arrange the following alkanes in order of decreasing boiling point. Explain your answer. 6m

2,2-dimethylbutane, hexane, butane

Answer

Hexane, 2,2-dimethylbutane, butane

Boiling point increases as **the number of carbon increases @ molecular mass @ molecular weight** because the strength of van der Waals forces between molecules increases.

For alkane isomers, **branching decreases the contact surface area** which decreases the strength of van der Waals, thus decreases the boiling point.

Q. 3 (b) Compound **A**, C_7H_{14}, contains a five-membered ring and two methyl groups as substituents. 4 m

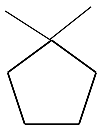
(i) Draw all structural isomers of A.

(ii) Which isomer can exist as geometrical isomer?

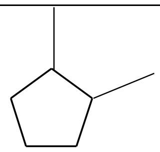
Draw the structures and explain your answer.

Answer (b)

(i)



I



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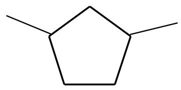


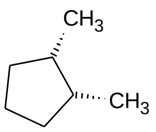
III

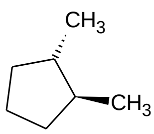
Note : 2M – 3 correct structures

1M – 2 correct structures

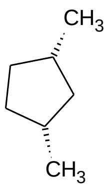
II or III

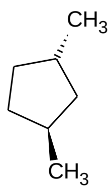
(ii)





OR





Explanation :

Both methyl either at same or opposite side

Q. 3 (c) Reaction of 2-methylpropane with chlorine gas in the presence of ultraviolet light forms two monochlorinated products. 5 m

(i) Draw the structures of the products.

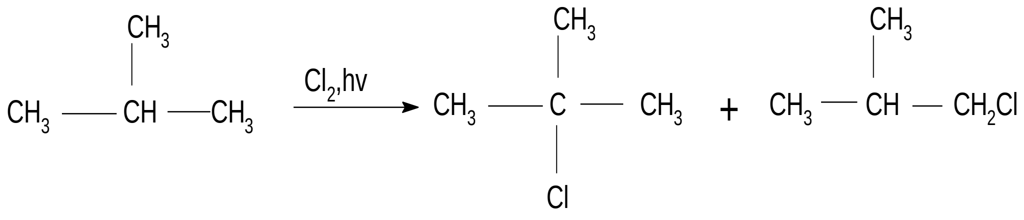
(ii) Name the mechanism for the formation of the products.

(iii) Draw the most stable free radical formed in this reaction.

(iv) Write reaction equations of the steps for the formation of major product.

Answer (c)

(i)



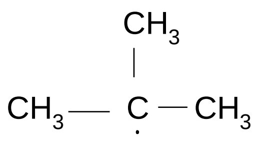
Note :

1M – for each product

(ii)

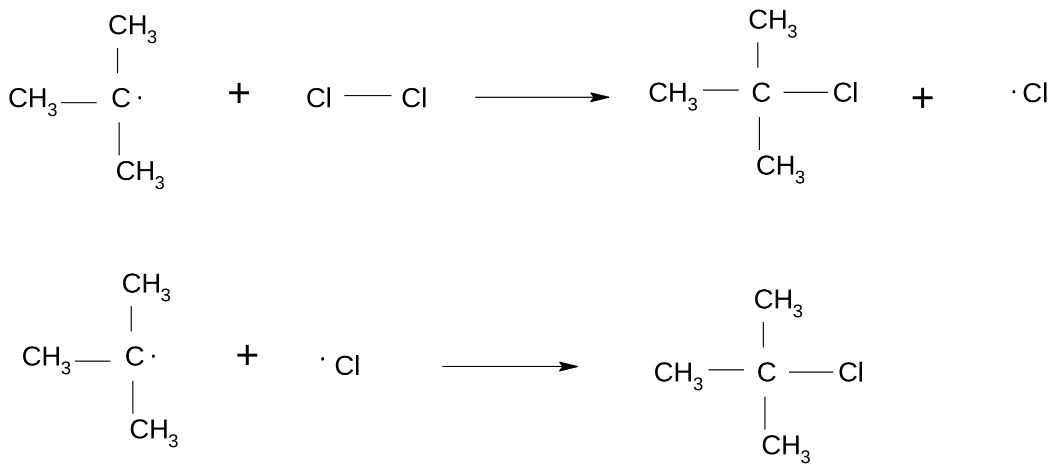
Free radical substitution

(iii)



Note : 1 M – for each step

(iv)



Q. 4

Hydrocarbon **P**, C_4H_8, has five structural isomers **Q, R, S, T** and **U**. isomers **Q, R**, and **S** react with bromine in CH_2Cl_2 to produce the respective alkyl halides but **T** and **U** do not. The boiling points of isomers increase in the order of **S, Q**, and **R**. Two isomers react with HCl to produce an optically active product.

Draw the the structures of **Q, R, S, T**, and **U** and explain your answer. Indicate the isomers that give optically active product, write the chemical equation and draw the mechanism involved.

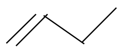
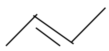
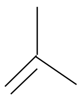
Explain why **S** has the lowest boiling point.

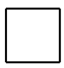
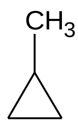
Suggest a chemical test other than bromine in CH_2Cl_2 that can be used to differentiate **Q** from **T**. state the observation and write the chemical equation involved.

Answer

Reaction with Br_2/ CH_2C_{l2} = Q,R and S are **alkenes @ unsaturated HC**

No reaction with Br_2/ CH_2Cl_2 = T and U are **alkane @ saturated hydrocarbon / cycloalkane**

Q =  R =  S = 

T =  U = 

Note: Q can be exchanged with R
T can be exchange with U

Isomers produced optically active compound = Q and R

Equation:

$CH_2{=}CHCH_2CH_3 + HCl \longrightarrow CH_2(Cl)CH_2CH_2CH_3$

Optically active product

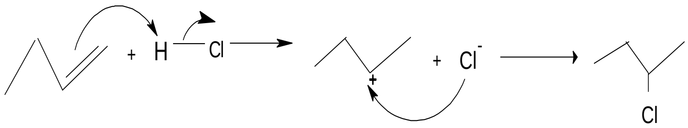
OR

$CH_3CH{=}CHCH_3 + HCl \longrightarrow CH_3CH(Cl)CH_2CH_3$

Optically active product

Note: 1M – Product
1M –Overall equation

Mechanism



1M 1M 1M 1M

Note: 1M for each arrow and 1M for carbocation

S has lowest boiling point because it is a **branched alkene** and therefore has **low surface area / lower contact area / smaller surface area**

Test:

Bromine water*OR***alkaline $KMnO_{4}$, cold (Baeyer's test)**

If use Bromine water,

Observation: Q decolourised bromine water

T no observable change

Equation:

$CH_2{=}CH_2CH_2CH_3 + Br_2/H_2O \longrightarrow CH_2(Br)CH(OH)CH_2CH_3$

OR

If use $KMnO_4$, cold/ Baeyer's Test

Observation: Q gives brown precipitate

T no observable change

Equation:

$CH_2{=}CH_2CH_2CH_3 + KMnO_4, OH^- \longrightarrow CH_2(OH)CH(OH)CH_2CH_3$

$+ MnO_2$ ↓

@ : Q turn from purple to colourless

T no observable change

Equation:

$CH_2{=}CH_2CH_2CH_3 + KMnO_4, OH^- \longrightarrow CH_2(OH)CH(OH)CH_2CH_3$

PSPM 2013 / 2014

Q. 5 Both 2-methylpropane and 2-methylpropene react with bromine gas in different reaction conditions to yield brominated products. 7 m

(a) Draw the structures of all possible products formed.

(b) Give the reaction condition for the formation of each product.

(c) Write chemical equations for all reactions involved.

(d) Propose a chemical test to differentiate 2-methylpropane from 2-methylpropene. State the observation and write the reaction equation.

(e) 2-methylpropene can be prepared from alcohols. Suggest the suitable alcohols for this preparation and name the type of reaction.

Answer

(a) 2-methylpropane: $(CH_3)_3C\text{-}Br$

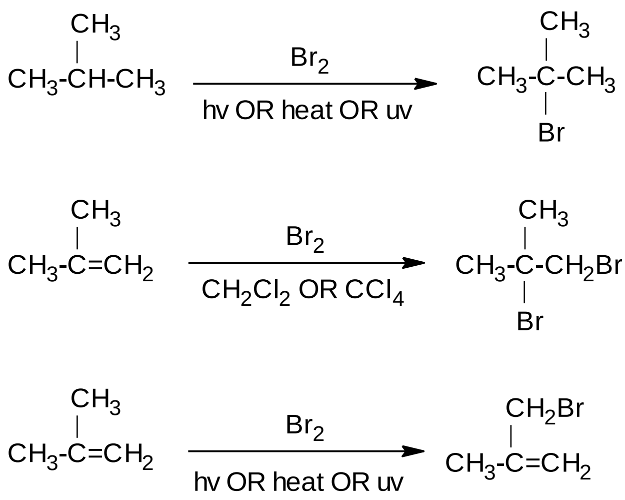
2-methylpropene: $(CH_3)_2CBrCH_2Br$, $CH_3(CH_2Br)C{=}CH_2$

(b) $(CH_3)_3C\text{-}Br$: Br_2, hv or heat or uv

$(CH_3)_2CBrCH_2Br$: Br_2, CH_2Cl_2 or CCl_4

$CH_3(CH_2Br)C{=}CH_2$: Br_2, hv or heat or uv

(c)



(d) Reagent: Bromine water or Br_2 in CH_2Cl_2 or $KMnO_4$, OH^-, cold

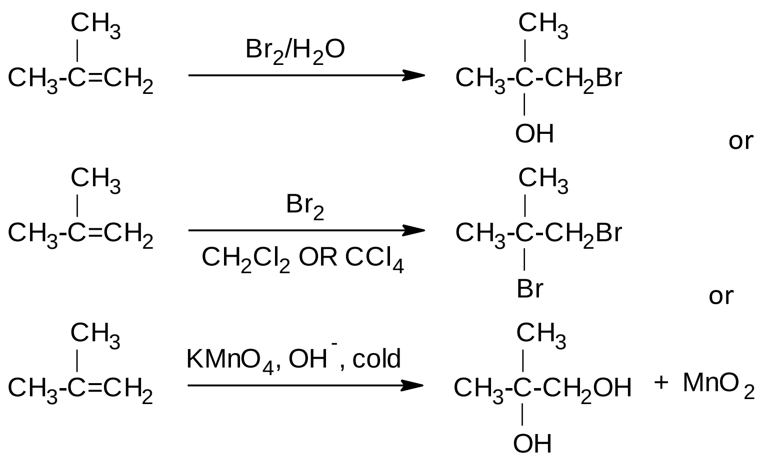
<u>Observation:</u>

2-methylpropene: **Decolourises** bromine water OR

Decolourises Br_2 in CH_2Cl_2 OR

Decolourises purple colour of $KMnO_4$ / brown precipiate

2-methylpropane: No observable changes with Bromine water or Br_2 in CH_2Cl_2 or $KMnO_4$, OH^-, cold



(e) $(CH_3)_2C(OH)CH_3$ or 2-methyl-2-propanol / 2-methylpropan-2-ol

$(CH_3)_2CHCH_2OH$ or 2-methyl-1-propanol / 2-methylpropan-1-ol

Type of reaction: Dehydration OR Eliminition

Q. 6 Hydrocarbon ***Q***, C_7H_{14} is an unsaturated optically active compound that exhibits geometrical isomerism. Compound ***Q*** reacts with reagent ***R*** to yield ***S***. ***S*** forms a cloudy solution immediately with Lucas reagent. 8 m

Treatment of ***Q*** with reagent ***T*** produces a mixture of ***U*** and ***V***. Compounds ***U*** and ***V*** give fruity smell when reacted with ethanol in the presence of traces of mineral acid.

State reagents ***R*** and ***T***. Deduce the structures of ***Q***, ***S***, ***U*** and ***V***. Draw all possible stereoisomers of ***Q***. Give the reaction mechanism for the formation of ***S***. Write the reaction equation when ***U*** or ***V*** reacts with ethanol.

Answer

Reagents :

R = H_2O, H^+

T = $KMnO_4$, H^+, Δ

Q: $CH_3CH{=}CHCH(CH_3)CH_2CH_3$

S : $CH_3CH_2CH_2C(OH)(CH_3)CH_2CH_3$

U: CH_3COOH

V: $HOOCCH(CH_3)CH_2CH_3$

Note : **U** and **V** are interchangeable

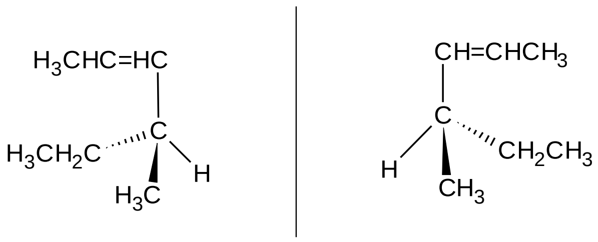
Q is an alkene that has a **chiral centre**

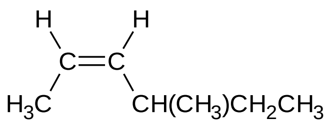
Has two same substituents attached to C=C

or **can exist as cis and trans isomers**

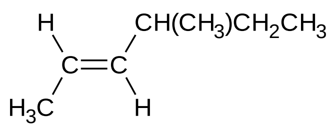
S is a 3° alcohol

U and V are carboxylic acids

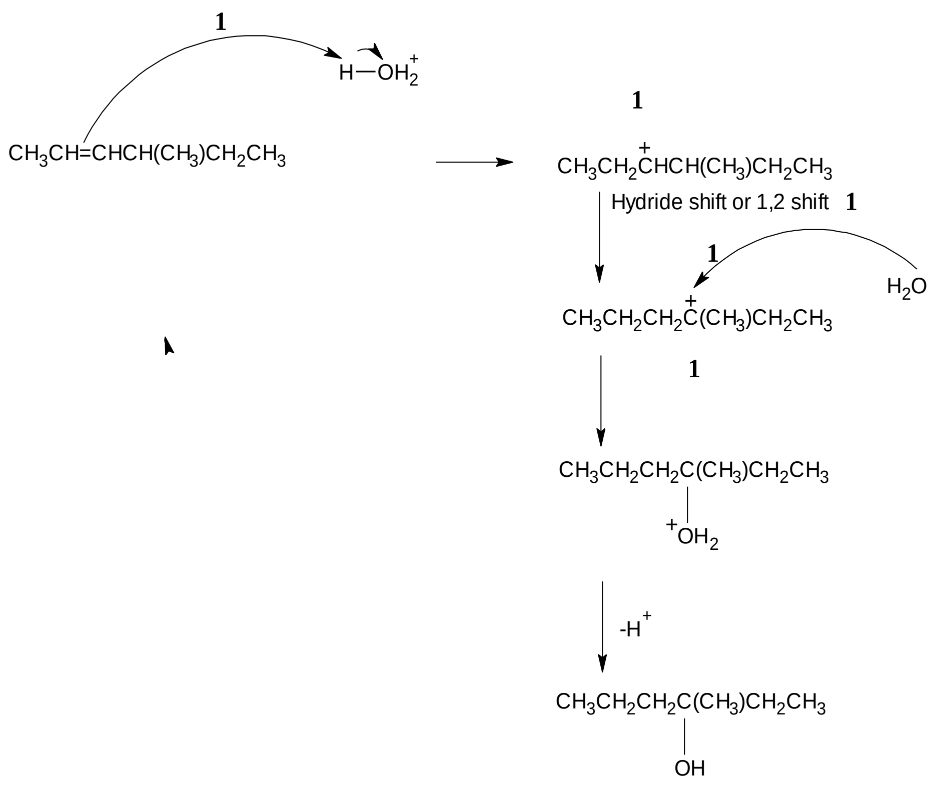




cis



trans



$$CH_3COOH + CH_3CH_2OH \xrightarrow{H^+} CH_3COOCH_2CH_3 + H_2O$$

OR

$$HOOCCH(CH_3)CH_2CH_3 + CH_3CH_2OH \xrightarrow{H^+} CH_3CH_2CH(CH_3)COOCH_2CH_3 + H_2O$$

PSPM 2014 / 2015

Q. 7 (a) Arrange the following alkanes in order of increasing boiling points. Explain your answer. 7 m

2,2-dimethylheptane, 4-ethylheptane, 2,2,4,4-tetramethylpentane and nonane

Answer (a) 2,2,4,4-tetramethylpentane < 2,2-dimethylheptane < 4-ethylheptane < nonane
Nonane – **straight chain / unbranched**, larger surface area, the strength of **van der Waals forces increases**, highest boiling point.

2,2,4,4-tetramethylpentane – **Branches increase**, smaller surface area, molecule more compact, the strength of **van der Waals forces decreases,** lowest boiling point.

Q. 7 (b) Monobromination of propane gives a mixture of products.

(i) Write a chemical equation for this reaction.
(ii) Determine the major product.
(iii) Write the reaction mechanism involving initiation, propagation and termination steps for the formation of the major product.

Answer (b)

(i)

$$CH_3CH_2CH_3 + Br_2 \xrightarrow{uv\ @\ hv} CH_3\underset{\substack{|\\ Br}}{C}HCH_3 + CH_3CH_2CH_2Br$$

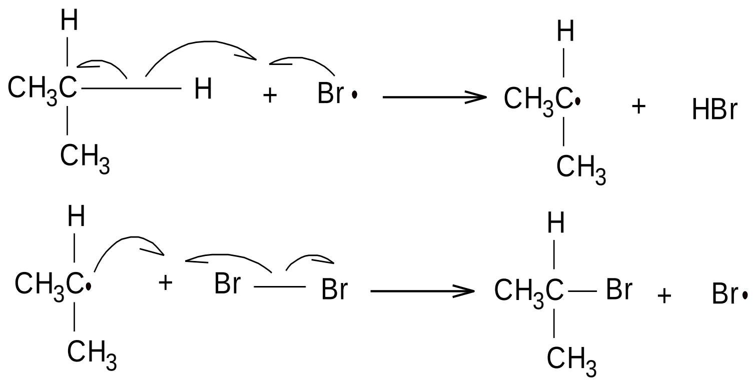
(ii)

$$CH_3\underset{\substack{|\\ Br}}{C}HCH_3 \quad @ \quad \text{2 – bromopropane}$$

(iii) Initiation step:

$$Br—Br \xrightarrow{uv} 2Br\bullet$$

Propagation step:



Q. 7 (c) Combustion of alkanes in air gives non-poisonous and non-polluting products, depending on the condition in which the combustion takes place. Write the chemical equations when 2-methylpentane reacts in 8 m

(i) excess oxygen
(ii) limited oxygen

Answer (c)

(i)

$$CH_3\underset{\substack{|\\ CH_3}}{C}HCH_2CH_2CH_3 + 19/2\ O_2 \longrightarrow 6CO_2 + 7H_2O$$

(ii)

$$CH_3\underset{\substack{|\\ CH_3}}{C}HCH_2CH_2CH_3 + 13/2\ O_2 \longrightarrow 6CO + 7H_2O$$

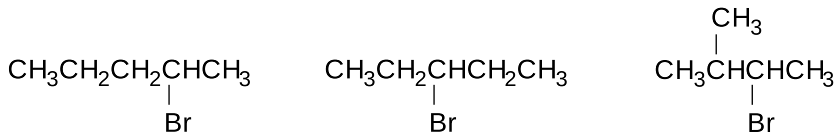
Q.8 An alkyl halide with molecular formula $C_5H_{11}Br$ exists in various isomeric forms. Draw **three** isomers of secondary alkyl halide. These three isomers undergo dehydrohalogenation reactions to give four products ***K***, ***L***, ***M*** and ***N***. Draw the structures of these products. Identify which product can exist as geometrical isomers. Explain.

Ozonolysis of ***K*** produces propanone and C_2H_4O. Write the chemical equations for this reaction. Treatment of ***L*** with hydrogen bromide gives a mixture of ***P, Q*** and ***R***. Draw the structures of ***P, Q*** and ***R***. Label the major product and write the mechanism for its formation.

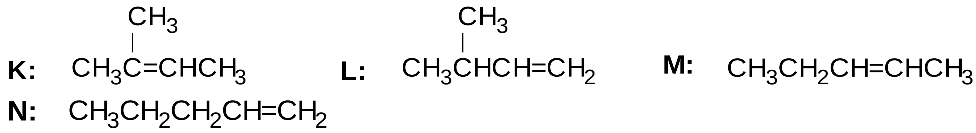
Suggest the product formed when ***L*** is reacted with hydrogen bromide in the presence of hydrogen peroxide.

Answer

Isomers of secondary alkyl halide:



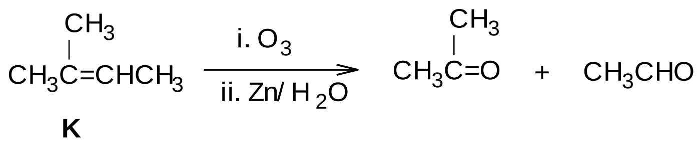
Products of dehydrohalogenation reaction:

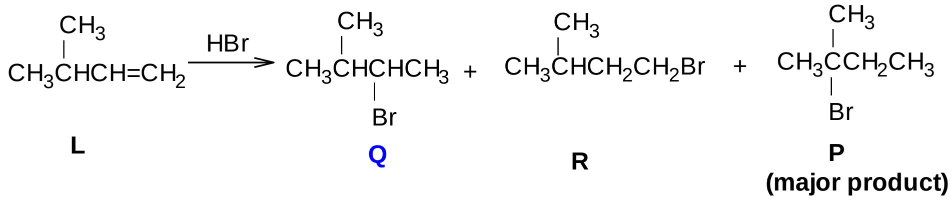


Product exists as geometrical isomer: ***M***

Reason: ***M*** has different groups of atoms at each C=C

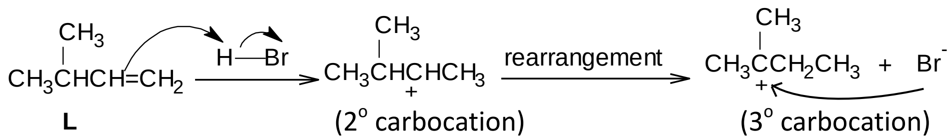
Ozonolysis of ***K***:



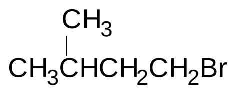


Note: ***Q***, ***R*** and ***P*** are interchangeable

Mechanism:



Product of reaction with HBr/H_2O_2



PSPM 2015 / 2016

Q. 9 (a) 2-methylbutane reacts with bromine in presence of UV light to give monobrominated product. 8 m

(i) Draw the structure of the major product formed.

(ii) Name the type of reaction.

(iii) Give the reaction mechanism for (a)(i).

Answer (a)

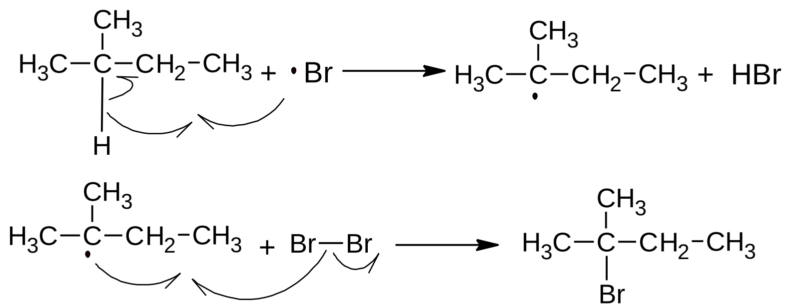
(i) $(CH_3)_2CBrCH_2CH_3$

(ii) Free radical substitution

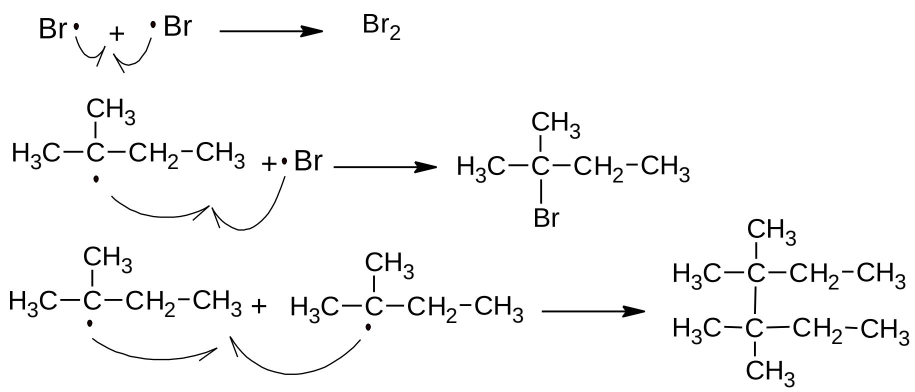
(iii) Initiation step:

$$Br - Br \xrightarrow{uv} 2Br\bullet$$

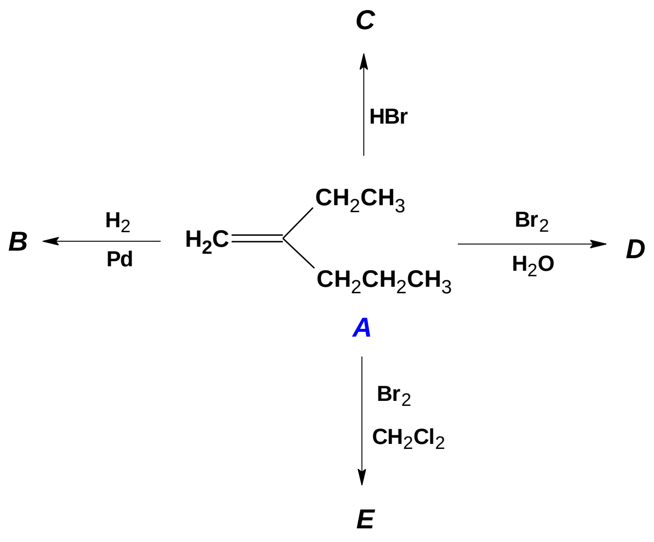
Propagation Step



Termination Step



Q. 9 (b) Reaction of 2-ethyl-1-pentene with seversl reagents are shown below: 7 m



(i) Draw the structures of B, C, D and E

Answer (b) (i)

B : $CH_3CH(CH_2CH_3)CH_2CH_2CH_3$

C : $CH_3CBr(CH_2CH_3)CH_2CH_2CH_3$

D : $Br{-}CH_2C(CH_2CH_3)(OH)CH_2CH_2CH_3$

E : $Br{-}CH_2C(CH_2CH_3)(Br)CH_2CH_2CH_3$

(ii) Suggest an alkyl halide and suitable reaction condition to produce A.

(ii) Alkyl halide : Cl @ Br

$Br{-}CH_3CHCH(CH_2CH_3)CH_2CH_2CH_3$

@

$CH_3CBr(CH_2CH_3)CH_2CH_2CH_3$

@

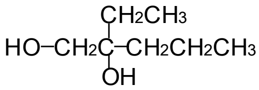
$CH_3CH(CH_2CH_3)CHBrCH_2CH_3$

$CH_3C(CHBrCH_3)(H)CH_2CH_2CH_3$ @

(iii) Reaction of A with cold dilute alkaline KMnO4 solution can be used as a confirmatory test for an alkene. Name this test, state the observation and draw the structure of product.

(iii) Reagent : Ethanol / Ethanolic KOH @ Ethanolic NaOH @ NaH

Baeyer's test

Product : 

The decolourised of purple colour @ the appearance of a brown precipitate

Q 10 Acid catalyzed dehydration of 2-pentanol gives compound V and W. Compound V exhibits geometrical isomerism. Ozonolysis of compound V produces Z and AA. Draw the structures of V and W. identify the major product and explain your answer. Draw geometrical isomers of V.

Write the mechanism for the dehydration of 2-pentanol. Draw the structures of Z and AA. Write reaction equation for the ozonolysis of V.

By using a suitable alkyl halide and Z or AA, show how 2-pentanol can be synthesized via a Grignard reagent.

Answer

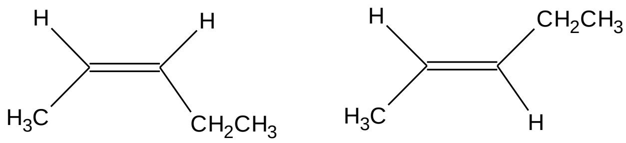
Structure of 2-pentanol

V : $CH_3CH{=}CHCH_2CH_3$

W : $CH_2{=}CHCH_2CH_2CH_3$

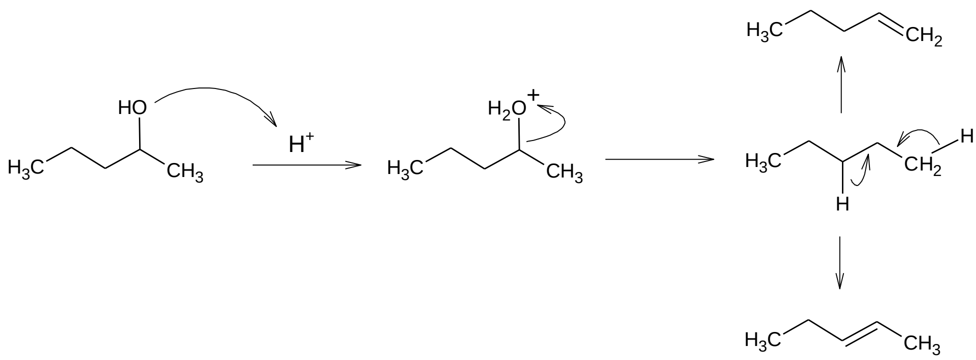
Major product : ***V,*** follow Saytzeff's rule.

Geometrical isomer :



cis **trans**

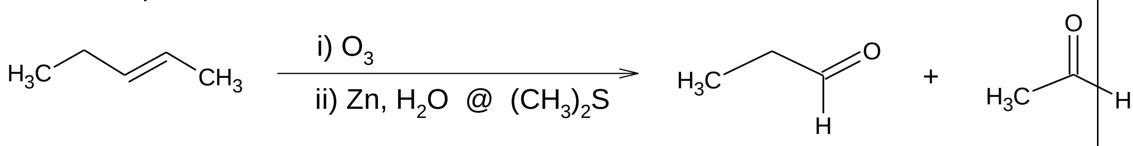
Mechanism :



Z : CH_3CH_2CHO @ CH_3CHO

AA : CH_3CHO @ CH_3CH_2CHO

Reaction equation :



$$CH_3CH_2CH_2X \xrightarrow[\text{Dry ether}]{\text{Mg}} CH_3CH_2CH_2MgX \xrightarrow[\text{ii) } H_3O^+]{\text{i) } CH_3CHO} CH_3CH_2CH_2CH(OH)CH_3$$

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Q. 11 (a) D is the isomer of alkane with molecular formula C_6H_{14} which possesses a quaternary carbon. The reaction scheme that involves D is shown below. 6 m

$$\underset{B}{C_6H_{14}O} \xrightarrow[\Delta]{\text{conc.}H_2SO_4} \underset{C}{C_6H_{12}} \xrightarrow[\text{Pd}]{H_2} \underset{D}{C_6H_{14}} \xrightarrow[\text{uv}]{Br_2} \underset{\text{(major)}}{E}$$

(i) Dehydration of secondary alcohol B produces compound C. Suggest the structure of B and write the mechanism of this transformation.

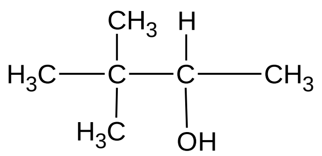
(ii) Suggest the structural formulae for C and D.

(iii) E is the major product of the reaction of D with Br2 in the presence of UV. Draw and classify the structure of E.

(iv) Suggest a chemical test to distinguish between C and D. Write the chemical equations and state your observations.

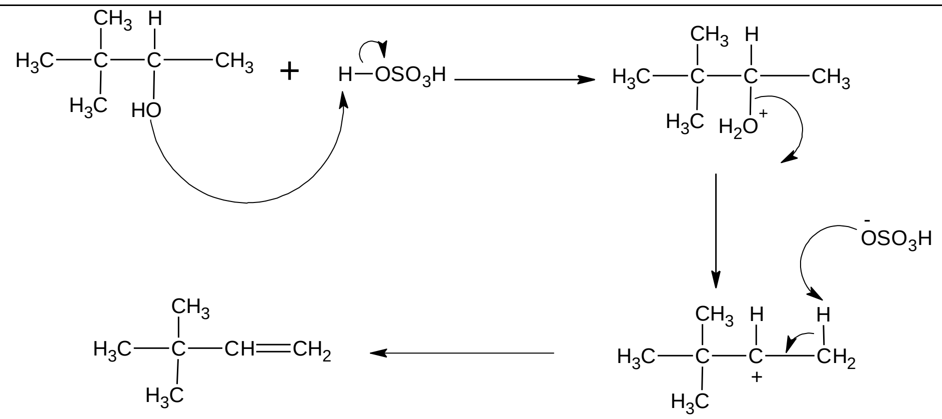
Answer (a)

(i)



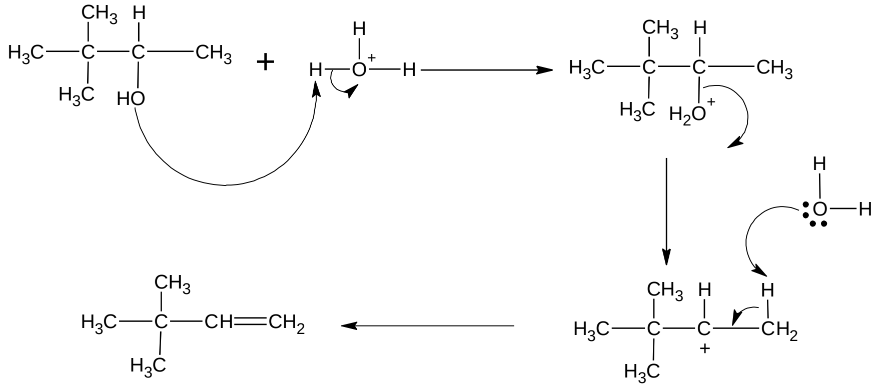
B

Mechanism :



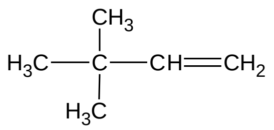
(ii)

OR

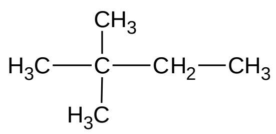


(iii)

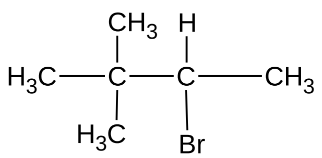
(iv)



C



D



E

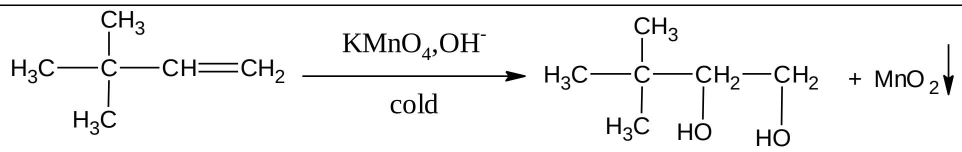
2°Haloalkane / alkyl halide / bromoalkane

Chemical test : Baeyer's Test OR Bromine Test

Chemical equation and observations:

<u>Baeyer's Test</u>

Alkene



Observation: Purple colour of $KMnO_4$decolourized OR brown precipitate of MnO_2 is formed.

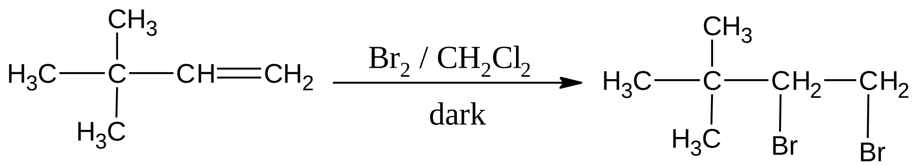
Alkane

Observation : Purple colour of KMnO4 remains unchanged.

OR

<u>Bromine Test</u>

Alkene



Observation: Reddish browncolour of bromineturn to colourless (decolourized).

Alkane

Observation :Reddish browncolour of bromine remains unchanged.

Q. 12 (a) State **two (2)** types of bond cleavage that can occur in a covalent bond. Explain how these cleavage occur and identify the species formed. 5 m

Complete the following reaction equation by showing the movement of electrons using curved arrows. Discuss the bond cleavage involved in the reaction.

$$H—Cl \ + \ :NH_3 \ \rightarrow$$

Answer

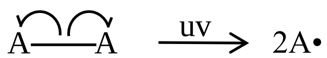
Homolytic

Heterolytic

<u>Homolytic cleavage</u>

Two electrons that are involved in the bond are **distributed equally** generating **free radicals**.

OR



<u>Heterolytic cleavage</u>

Two electrons that are involved in the bond are **distributed to the more electronegative species** generating a pair of **oppositely charged ions/ cation and anion/ positive ion and negative ion**.

OR

$$A-B \longrightarrow A^+ + B^-$$

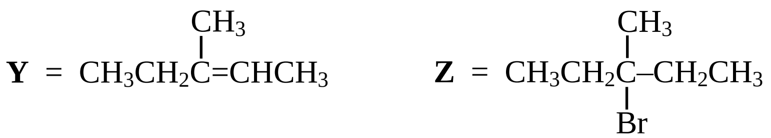
$$H-Cl + :NH_3 \longrightarrow NH_4^+ + Cl^-$$

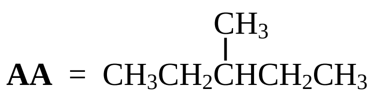
<u>Heterolytic cleavage</u>
Cl is more electronegative than H.
Both electrons in the bond attracted towards Cl, leaving H^+ ion @ forming Cl^- and NH_4^+.
NH_3 is a nucleophile @ has lone pair that attacks H^+ ion.

Q. 12 (b) Alkenes are unsaturated hydrocarbons and chemically more reactive than alkane. Treatment of an alkene, **Y** with halogen acid, HBr, gives **Z** while hydrogenation of **Y** forms **AA**. Reaction of **Y** with reagent **BB** gives butanone and ethanoic acid as the products. Explain the reactivity of **Y** and determine the type of reaction involved. Draw the structures of **Y**, **Z**, **AA** and identify reagent **BB**. Show the isomerism of **Y**. 4 m

Answer

Reactivity of Y is due to the **presence of carbon-carbon double bond / C=C bond / double bond of alkene**
Type of reaction : **Addition reaction**





BB = $KMnO_4$, H^+, Δ

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Q. 13 Reaction of reactive bromine radical with compound **A** forms a monobrominated product **B**, while compound **C** reacts with bromine gas in dicholoromethane to produce compound **D**. 9 m

Compound **A**	3-tert-butylhexane
Compound **C**	3-tert-butyl-3-hexene

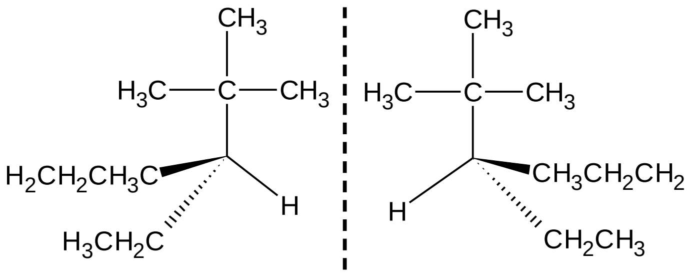
(a) Draw the structural formulae for compounds **A** and **C**.

(b) Identify which of the compounds in 2(a) exhibits chirality. Draw a pair of enantiomer using 3-dimensional formulae for identified compound.

Answer (a) Compound **A** : $CH_3CH_2CH_2CHC(CH_3)_3CH_2CH_3$

Compound **C** : $CH_3CH_2CH{=}CC(CH_3)_3CH_2CH_3$

(b) Compound **A**



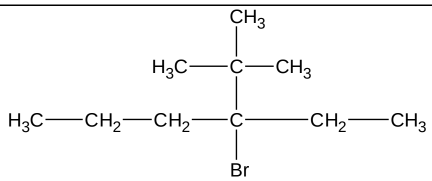
(c) Name the type of reaction involved in the formation of **B** and **D**.

Answer (c) Compound **B** : Free radical substitution reaction

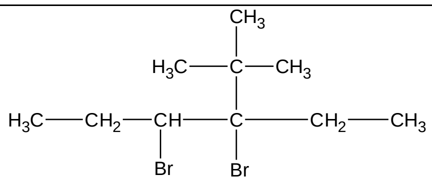
Compound **D** : Electrophilic addition reaction

(d) Draw the structures of **B** and **D**.

Answer (d)



B



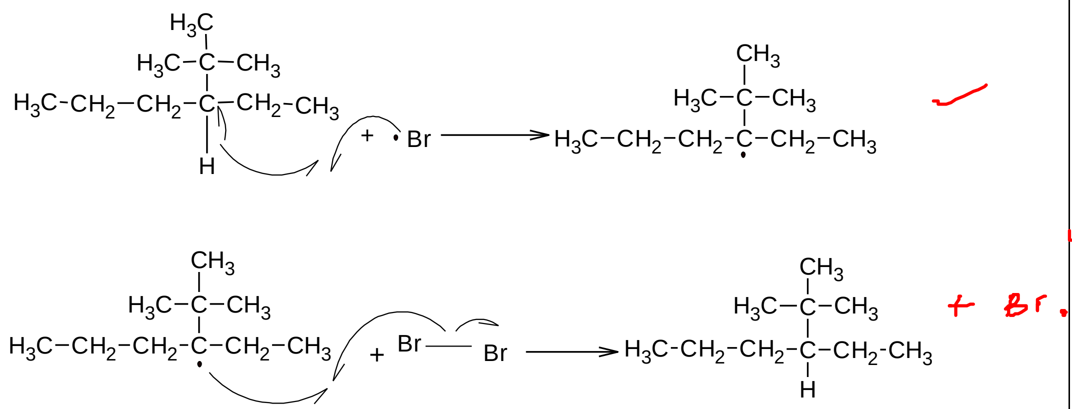
D

(e) By using curved arrows show the initiation and propagation steps for the formation of **B**.

Answer (e) Initiation step:

$$Br-Br \xrightarrow{uv} 2Br\cdot$$

Propagation step:



(f) Suggest a chemical test to distinguish between **A** and **C**. State the observations and write the chemical equation involved.

Answer (f) Baeyer's Test / cold dilute alkaline $KMnO_4$

Observation :

Compound **C** : Purple colour decolourised @ brown precipitate formed.

Compound **A** : No observable change @ purple colour remain unchanged.

$$CH_3CH_2CH{=}CC(CH_3)_3CH_2CH_3 \xrightarrow[cold]{KMnO_4,\ OH^-} CH_3CH_2CH(OH)C(OH)C(CH_3)_3CH_2CH_3 + MnO_2\downarrow$$

@ ~~Bromine water~~ @ Bromine in CH_2Cl_2 @ Bromine test in the dark

Compound **C** : Reddish brown colour decolourised

Compound **A** : No observable change @ reddish brown colour remain unchanged.

$$CH_3CH_2CH{=}CC(CH_3)_3CH_2CH_3 \xrightarrow[\text{in the dark}]{Br_2/\cancel{H_2O}\ CH_2Cl_2} CH_3CH_2CH(Br)C(OH)C(CH_3)_3CH_2CH_3$$

Eg A

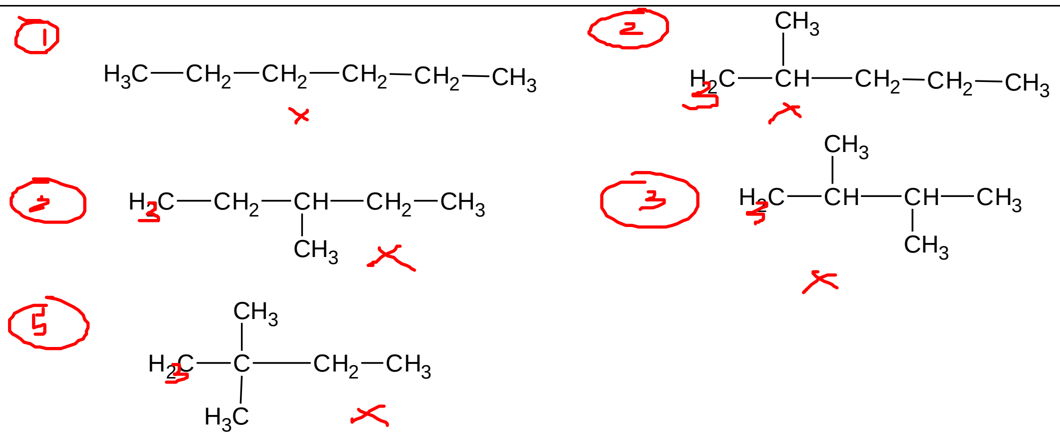
@

$$CH_3CH_2CH{=}CC(CH_3)_3CH_2CH_3 \xrightarrow[\text{in the dark}]{Br_2/CH_2Cl_2} CH_3CH_2CH(Br)C(Br)C(CH_3)_3CH_2CH_3$$

Q. 14 (a) Hexane, $\cancel{C_6H_6}$ C_6H_{14}, can exist in many isomeric forms. Draw all structural isomers of hexane. Is there any chiral carbon? Explain your answer.

Name the type of bond cleavage when isomers of hexane undergo halogenation reaction in the presence of light. Write an equation to show this bond cleavage. 6 m

Answer



No chiral isomer.

N chiral center @ no carbon that hold 4 different groups @ no stereogenic centre

Bond cleavage : Homolytic

Equation :

$$\cdot X + C_6H_{14} \xrightarrow{uv} R\cdot + HX$$

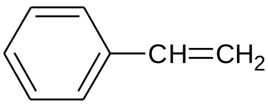
@ any other possibilities

$$CH_3CH_2CH_2CH_2CH_2CH_3 + X_2 \xrightarrow{uv} CH_3CH_2CH(Cl)CH_2CH_3 + HCl$$

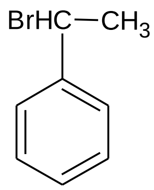
Q. 14 (b) Hydrocarbon **S**, C_8H_8, reacts with hydrogen in the presence of platinum catalyst to yield **T**, C_8H_{10}. Compound **U** is formed when **T** is heated with alkaline potassium permanganate solution, followed by hydrolysis. The reaction of **S** with hydrogen bromide gives **V**, whereas ozonolysis of **S** produces **W** and methanal, $H_2C{=}O$. Draw the structures of **S** to **W**. Give the name of the reaction for the conversion of **S** to **T**. Write all chemical equations involved.

Answer

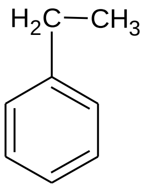
S :



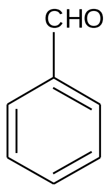
V :



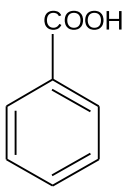
T :



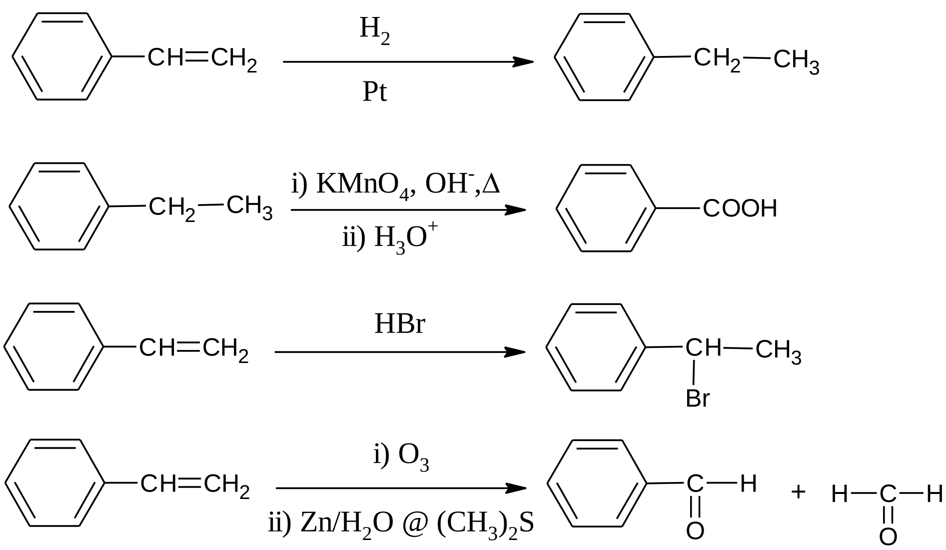
W :



U :



Hydrogenation



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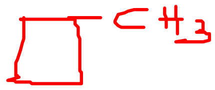
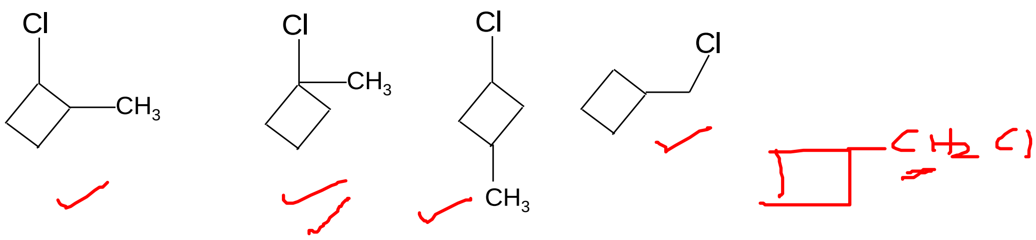


4C

Q. 15 Compound A is a four-membered ring alkane with molecular formula C_5H_{10}. 7m

(a) Draw the structural formulae for all possible monosubstituted products for the reaction between C_5H_{10} and chlorine in the presence of light.

Answer

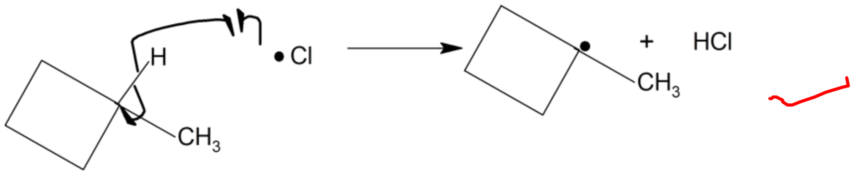


(b) Write the reaction mechanism for the formation of the most stable reaction intermediate in **4(a).**

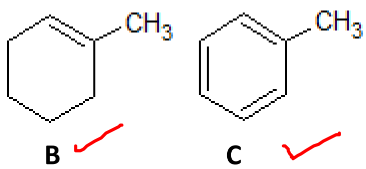
Answer 1

$$Cl-Cl \xrightarrow{uv} 2\bullet Cl$$

2



Q. 16 Compounds **B** and **C** are hydrocarbons with the structural formulae as shown below. 8m



(a) Name compounds **B** and **C** according to IUPAC nomenclature.

Answer

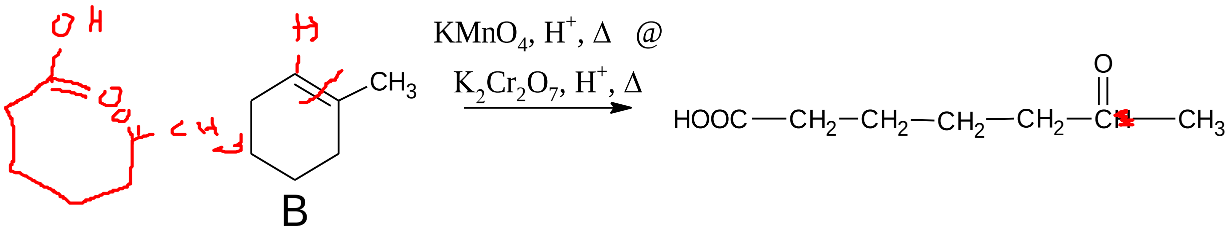
B : 1-methylcyclohexene @ methylcyclohexene

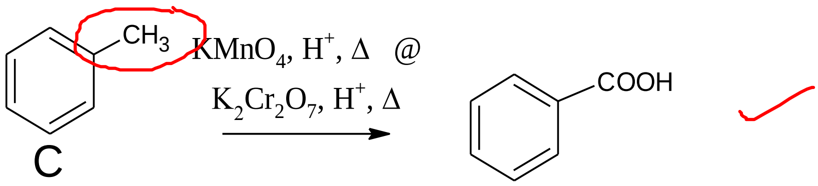
C : methylbenzene@ toluene

Note: **1-methylbenzene not accepted

(b) Both **B** and **C** can undergo oxidation reaction with the same oxidizing agent. Write chemical equations involved and explain the differences between these two reactions.

Answer





B : Oxidative cleavage/ double bond breaking

C : Oxidation at the side chain/ the presence of benzylic hydorogen

(c) Name one reaction that converts **B** to methylcyclohexane

Answer

Reduction

@ hydrogenation